Hypergraph-Guided Multimodal Prototype for Remote Sensing Scene Understanding

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***Abstract*— Noticeable achievements have been made in entity-level perception tasks (e.g., object detection) in remote sensing (RS) image interpretation. But for RS images carrying rich content, individual perception cannot well obtain the interaction patterns between entities. The recognition of relationships between entities is the key to deeply understanding RS scenes. In this article, we propose a hypergraph-guided multimodal prototype network (HMPNet), which performs relation recognition by matching relation representations with multimodal predicate prototypes. To overcome the imbalance of modal information in the matching process, a multimodal calibration strategy is devised, taking into account the image subprototype and text subprototype, which makes prediction results more reliable. Meanwhile, to align image and text subprototypes and explore relevant semantic patterns, the multimodal hypergraph is constructed to efficiently capture the associations between heterogeneous prototypes. Experimental results show that the performance of our model can reach the state-of-the-art (SOTA) level on the RS scene graph generation (SGG) task.**

***Index Terms*— Hypergraph network, multimodal prototypes, remote sensing (RS) scene understanding, scene graph generation (SGG).**

I. Introduction

With the advancement of modern remote sensing (RS) technology, a large number of high-quality RS images are accessible [1], [2], [3], [4], [5], and the requirements for RS image interpretation [3], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16] become more complex and changeable.

In recent works of image interpretation, researchers focus more on object detection [4], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27] and semantic segmentation [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], which aims to perform prediction for each entity or pixel. These interpretation tasks involve the individual perception of entities or regions but pay less attention to high-level cognition of relations between entities. This may lead to an insufficient understanding of the RS scene with rich content. As shown in Fig. 1(a) and (b), the semantic difference (whether the cargo ship is being towed or hoisted) cannot be distinguished by object detection because both the two images contain the same targets, i.e., dry cargo ships, tugboats, and cranes. It is clear that semantic relations between entity pairs are significant for truly understanding the evident messages of RS scenes. The relationship between pairs of entities is usually represented by a triplet, i.e., ⟨subject–predicate–object⟩. Accordingly, the main relations in Fig. 1(a) can be represented as ⟨tugboat#2–tow–dry cargo ship#1⟩ and ⟨tugboat#3–tow–dry cargo ship#1⟩. To explore complex relationships and achieve a further semantic understanding of images, the scene graph [38], [39] is proposed to effectively model the interactions between entities. The scene graphs of Fig. 1(a) and (b) are illustrated in Fig. 1(c) and (d). Therefore, through the indicative relation links between entity nodes, a more accurate higher level semantic understanding of the specific scene can be achieved.

Scene graph generation (SGG) is to generate a semantic graph structure as a representation of an input image. The nodes of the scene graph correspond to the target entities in the image, and the edges of the graph correspond to the relationships of pairs of entities. That is, the task is not only to detect entity nodes and predict their categories but also to classify the relationships of entity pairs for high-level scene cognition. For natural scenes, researchers have constructed the large-scale dataset [40], and some methods to generate natural scene graphs have emerged [41], [42], [43], [44] and achieved effective results. However, there is a notable scarcity of works aimed at enhancing higher level understanding in RS scenes. In fact, generating scene graphs for RS images presents greater challenges compared with natural scenes. Specifically, RS images contain more intricate and diverse information than natural images. This complexity arises because RS images usually contain a bird's-eye overview of a scene with a large spatial range, while natural images tend to be closed-ups of prominent parts of space. Consequently, the number of entities on each RS image is larger and the semantic relationship

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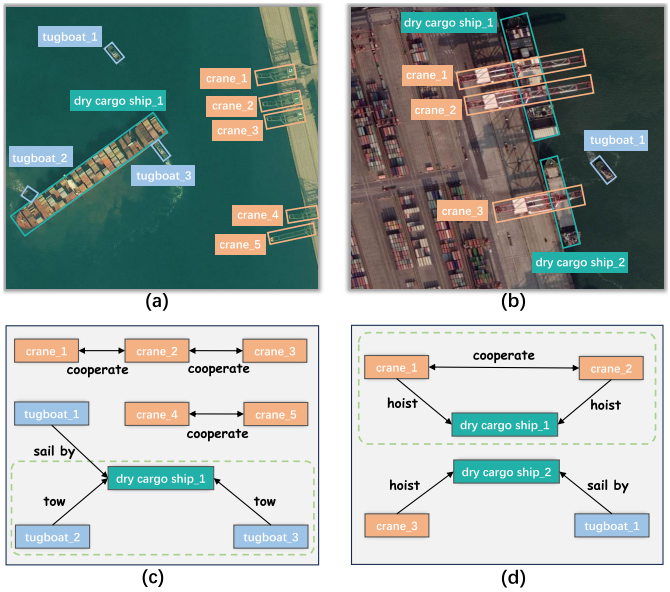


Fig. 1. Illustration of individual perception and relational cognition. Through entity-level detection, we can only perceive the existence of similar entities in scenes (a) and (b) and cannot understand their detailed semantic difference. By constructing scene graphs (c) and (d), interactions between entities are perceived by modeling semantic relationships of entity pairs. Accordingly, we can understand the important semantic contents contained in the RS scenes; that is, the dry cargo ship in (a) is towed by two tugboats, and the dry cargo ships in (b) are being hoisted with cargo by cranes.

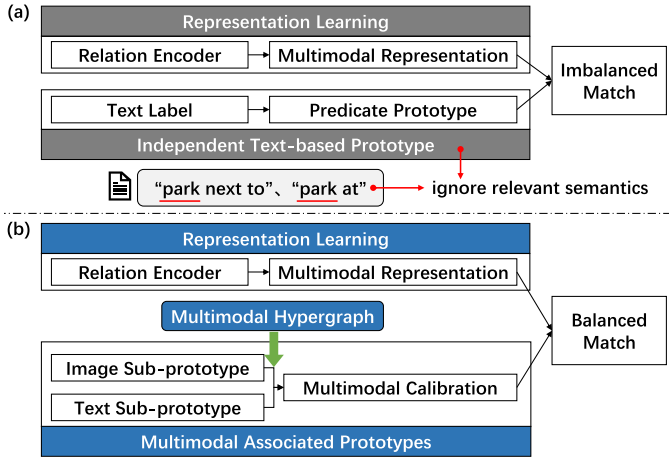


Fig. 2. Comparison of different prototype-based pipelines. (a) Previous method encodes text-based prototypes to match predicate representations, resulting in an imbalance problem during matching, and independent prototypes ignore the relevant semantic patterns between predicates. Our proposed HMPNet adopts a multimodal calibration to achieve more reliable and balanced matching. Furthermore, the multimodal hypergraph is devised to capture semantic associations while catalyzing image and text subprototypes.

interactions are more entangled. Accordingly, for the RS SGG task, the model is required to have a stronger discriminative ability for samples of various relationship categories. In addition, the semantics carried by more diverse predicate categories in RS scenes is also crucial, and how to model their inherent associations needs to be considered. To address this need, in this article, we propose a novel approach that fully leverages multimodal context and semantic associations to optimize remote sensing SGG by lifting the reliability of relationship prediction.

The key to the SGG task is how to model the entity/predicate representations to achieve pairwise relation recognition. Some common methods [42], [45], [46], [47] are to obtain predicate representations by fusing the features of entity pairs extracted by detectors and then feeding them into classifiers to predict relation labels. Later, to further compact entity/predicate representations and make them more precise, the prototype-based method [48] emerges, which encodes predicate categories as prototypes and performs relation recognition by matching relation representations and prototypes in the semantic space. Nonetheless, as shown in Fig. 2, there are some issues with this prototype-based method. First, the predicate representation in the previous method combines visual deep features from the input image and text features derived from entity labels. However, the predicate prototype is computed solely based on the text labels of the relation categories. Therefore, when matching predicate representation and prototype, the imbalance of modal information leads to suboptimal matching accuracy. Second, each predicate prototype is not semantically independent. For example, "park next to" and "park at" contain a strong relevant semantic pattern (i.e., "park"). Therefore, the independent optimization of these predicate prototypes will weaken their referentiality, resulting in lower effectiveness of the prototypes.

To conquer the above challenges, we propose a hypergraph-guided multimodal prototype network (HMPNet), which aims to achieve a more precise understanding of rich semantic content in remote sensing scenes. The comparison of the proposed model and the previous one is shown in Fig. 2. To solve the problem of modal information imbalance during matching, the multimodal calibration strategy is devised based on the image subprototype and the text subprototype, which injects aligned multimodal information into the matching process to achieve more reliable semantic relationship recognition. The multimodal calibration strategy fully considers the information encoded in both image and text modalities in relation representation. The proposed image subprototypes and text subprototypes are aligned and interacted to provide a comprehensive predicate prototype space that is more suitable for measuring multimodal relation representations and promotes a more balanced matching process. Meanwhile, to allow two modalities to interact with each other and capture relevant semantic patterns, the multimodal hypergraph is constructed to endow text and image subprototypes to purify themselves by absorbing complementary information. The hypergraph is relatively sparse in structure and efficient in passing messages, which transcends the local receptive field without introducing too much global noise. Also, it can effectively establish high-order correlations among multimodal prototypes. Under the mutual guidance and fusion association brought by hypergraphs, the calibrated representation-prototype matching process based on multimodal subprototypes is more reasonable and enjoys stronger discriminability.

In summary, our main contributions are four folds.

1) We propose the HMPNet, which identifies interaction relationships between entities by matching predicate representations with multimodal prototypes. This approach

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promotes a further advanced semantic understanding of remote sensing scenes.
2) To make the representation-prototype matching process more balanced, the multimodal calibration strategy is devised based on text and image subprototypes to boost the reliability of relation recognition.
3) The multimodal hypergraph is built to purify the contents of image and text subprototypes. Meanwhile, it can enhance the effectiveness of prototypes by mining and jointly optimizing their relevant semantic patterns.
4) Experimental results show that HMPNet can boost the accuracy of relation recognition, achieving SOTA performance on the remote sensing SGG task.

II. Related Works

A. General SGG

Most SGG models rely on fully connected layers for predicate recognition. The early method [39] uses separate visual and language modules to encode object and relation features, followed by a predicate classifier. After that, many works aim to enhance the encoding contents that represent objects and relations. Xu et al. [49] design an end-to-end model that iteratively updates the representations. MotifNet [41] computes global context by bidirectional long short-term memory networks (LSTMs). VCTree [42] perceives hierarchical relations by a dynamic object-level tree structure. Graph R-CNN [50] proposes the relation proposal network (RePN) to extract meaningful potential relations. Also, some debiasing models [44], [51], [52], [53] improve the accuracy by addressing imbalanced data.

In contrast, the prototype-based method [48] recognizes relations by matching more compact representations to encoded prototypes, which is more robust even in the face of long-tail data. As stated in Section I, this pipeline struggles with the issue of modal information imbalance, and the associations among predicates are not captured. To address these issues, we propose the multimodal prototypes coupled with the hypergraph to adapt to more complex remote sensing scenarios.

B. Remote Sensing Scene Understanding

Research on remote sensing scene understanding has progressed relatively slowly. To promote the development of relation cognition in the remote sensing field, Li et al. [54] develop RSICD [55] to a close-source remote sensing scene graph dataset. They propose a multiscale semantic fusion network to perceive semantic relationships between remote sensing entities. In addition to constructing the scene graph using bounding boxes of entities, Lin et al. [45] propose a novel segmentation-based SGG model that encodes the shapes of remote sensing entities. Lin et al. [56] use the attention-weighted context of entities and inject statistical knowledge of relations to simplify the search space for predicates. Note that the above remote sensing SGG models emphasize predicate representations and adopt conventional classifiers, which will be more difficult in complex remote sensing scenes. From another perspective, our HMPNet demonstrates that using hypergraphs to mine predicate associations can boost the performance of the relation prediction and the prototype pipeline is more effective in understanding remote sensing scenarios.

C. Hypergraph Learning

Beyond CNN models that use local receptive fields and grid sampling, researchers have explored graph-based networks [57], [58], [59], [60], [61], [62] to model data dependencies more efficiently, even for nongrid data. Feng et al. [63] propose hypergraph neural networks (HGNNs) that can flexibly model high-order correlation of data via hyperedge convolution. Bai et al. [64] further develop hypergraph convolution and attention operators to extend the applicability. Also, HGNN$^+$ [65] is a systematic hypergraph network framework. Jiang et al. [66] propose dynamic hypergraph neural networks (DHGNN) to dynamically update the hypergraph structure at each layer. Some researchers employ the hypergraph method to model the correlations among pixels of feature maps in the remote sensing segmentation task [28]. However, work using hypergraphs in remote sensing scene understanding tasks is rare. In this article, the multimodal hypergraph is an inspiring attempt at the prediction of higher level predicates, aiming to mine high-order correlations among heterogeneous subprototypes. Moreover, beyond information transfer within the hypergraph, we also explore the more efficient cross-modal message updating method via the hyperedges of multiple hypergraphs for multimodal data.

D. Multimodal Learning

Multimodal information fusion learning [67], [68], [69], [70] is an important research direction in the field of artificial intelligence. Some researchers integrate information from multiple modalities to achieve better data classification [71], [72]. The fusion operations in these methods primarily focus on combining images from different sources. In addition, fusion methods for image and text modalities have also been explored to enable cross-modal image retrieval [73], [74]. Furthermore, the multimodal contrastive learning approach, CLIP [75], is designed to build more powerful models using large-scale unlabeled Internet data. While these fusion methods have made significant progress in processing images and text sentences, they cannot directly handle fine-grained entities and predicates in scene graph tasks. In addition, nongrid multimodal subprototypes with high degrees of freedom impose new requirements on modeling methods. Accordingly, our HMPNet explores multimodal hypergraph learning to appropriately address these predicate-level problems.

III. Methodology

This section begins by presenting the overall framework of HMPNet. Next, it outlines the encoding process for subprototypes. Following this, the construction and reasoning methods of the multimodal hypergraph are discussed in detail. Finally, the calculation of the multimodal calibration strategy is explained, along with the definition of the total loss function.

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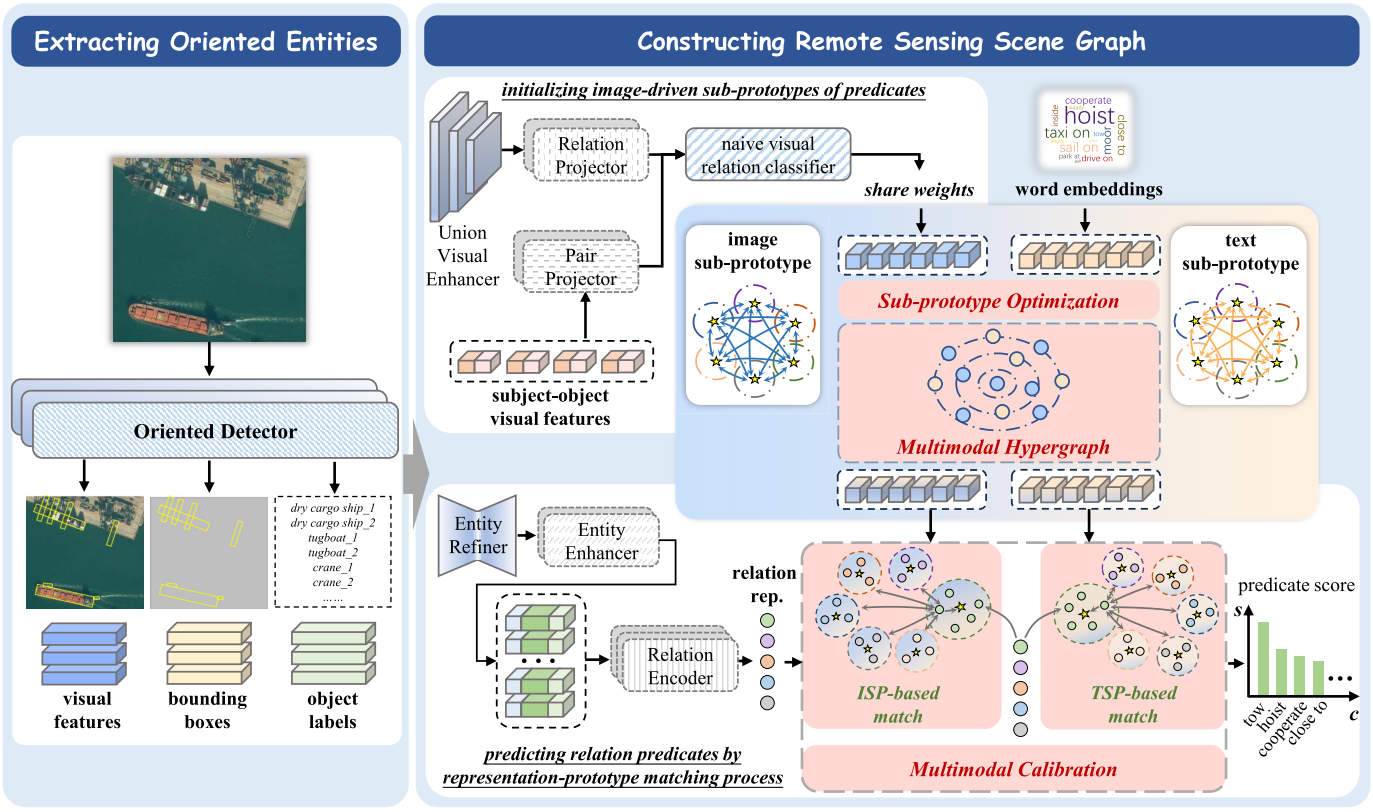


Fig. 3. Overall framework of the proposed HMPNet. The image is input into an oriented detector to extract entity information. After that, these elements are sent into the entity enhancer to obtain more powerful entity representations, and the relation encoder is used to generate predicate representations. A naive visual relation classifier is set to initialize the image subprototype. The SPO is performed during the training phase. Then, the image and text subprototypes are used to jointly perform multimodal hypergraph learning. Next, the ISP-based matching logits as well as TSP-based matching logits can be calculated. After conducting a multimodal calibrated representation-prototype matching process, the classification scores of relations are obtained.

A. Overall Framework

The proposed HMPNet is illustrated in Fig. 3. In the first stage, a pretrained detector extracts entity embeddings, including visual features, oriented bounding boxes, and class labels of proposal entities. In the second stage, a multimodal-prototype-based network performs relation recognition. A lightweight head, combining a union visual enhancer and relation projector, is designed to initialize the image subprototype and better capture predicate visual content. The text subprototype is initialized using word embedding. These image and text subprototypes are updated by performing cross-modal hypergraph reasoning. Then, the entity enhancer and relation encoder cooperate to extract the relation representation of hybrid modalities. The multimodal calibration is conducted to match representations to prototypes, balancing modalities and outputting the refined relation label.

B. Entity and Predicate Representations

Given an input image $\mathcal{I}$, a pretrained oriented detector is used to extract a set of oriented proposals for the entities existing on the image, which can be formulated as follows:

$$V, B, L, U = \text{OrientedDetector}(\mathcal{I}) \tag{1}$$

where V, B, and L denote collections of entity embeddings. To be specific, V is the set of visual deep features of entities, i.e., $V = \{v_i\}_{i=1}^{N}$, and $v_i \in \mathbb{R}^{4096}$. B is the set of oriented bounding boxes, i.e., $B = \{b_i\}_{i=1}^{N}$, and the coordinates of four corners of each oriented box are $b_i \in \mathbb{R}^8$. L is the set of entity labels, i.e., $L = \{l_i\}_{i=1}^{N}$ and $l_i \in \mathbb{R}^{C_e+1}$. Note that the number of entity classes is C_e and the number of selected proposals is N. In addittion, U is the set of visual deep features of union regions of entity pairs, i.e., $U = \{u_{ij} | i \neq j \& i, j \in N\}$, and $u_{ij} \in \mathbb{R}^{4096}$.

The class labels of subjects and objects play a crucial role in relation recognition. Therefore, the entity features obtained by the detector are fused to further refresh the entity labels. That is, the visual features, the spatial boxes, and the raw labels are jointly encoded to generate more accurate entity features and yield more reliable label predictions. This step is formulated as follows:

$$F = \text{RefineEncoder}([V; W_b B; \text{embed}(L)]) \tag{2}$$

where $F = \{f_i\}_{i=1}^{n}$ is the set of updated entity features. W_b denotes a fully connected layer to encode the coordinates of boxes. The pretrained language model Glove [76] is used to produce word representation, i.e., embed$(\cdot)$. $[\cdot;\cdot]$ is the concatenation operation

$$L^r = \text{RefineDecoder}(F) \tag{3}$$

where RefineDecoder is a fully connected layer and $L^r = \{l_i^r\}_{i=1}^{n}$ denotes the refined entity labels. Then, we can obtain

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the final entity representations $E = \{e_i\}_{i=1}^{n}$ by aggregating the above elements, the process is as follows:

$$E = \text{EntityEnhancer}\big(\big[V; F; \text{embed}\big(L^r\big)\big]\big). \tag{4}$$

Next, we can calculate each predicate representation r_{ij} using the following multi-layer perceptron (MLP) style relation encoder, which can be expressed as follows:

$$r_{ij} = \text{RelationEncoder}\big(\big[e_i; u_{ij}; e_j\big]\big) \tag{5}$$

where e_i and e_j are the specific pair of entities, corresponding to the ith subject and the jth object. u_{ij} is the relation visual features obtained in (1).

At this point, it is evident that the encoded predicate representations encompass multimodal information from both text and images. As a result, if prototype-based matching and classification are to be established, the predicate prototypes should also incorporate the corresponding multiple modalities.

C. Multimodal Subprototypes

The subprototype represents the center of the predicate semantics for a given modality. The relation sample, with its unique instance characteristics, should be closest in distance to the predicate prototype of the same category, which is fundamental to prototype-based relation recognition. In other words, the image and text subprototypes should be highly general and aligned to ensure consistency in the referential predicate semantics. With this in mind, we will delve into the design and optimization of the multimodal subprototypes.

1) Text Subprototype: For the text modality, we encode the predicate vocabulary to obtain the text subprototype. The ith text subprototype p_i^t is encoded using the word embedding of the relation label t_i. The process can be formulated as follows:

$$p_i^t = \text{MLP}^t(\text{embed}(t_i)) \tag{6}$$

where $\text{MLP}^t(\cdot)$ represents a two-hidden multilayer perceptron. embed$(\cdot)$ is employed to get the word embedding (Glove [76]) and $i \in \{1, 2, \ldots, C_r\}$. Note that C_r is the number of relation predicate categories.

2) Image Subprototype: For the image modality, we intend to extract an abstraction of the relation visual semantics to initially establish the image subprototype. To be specific, we apply a union visual enhancer (i.e., an MLP) to better perceive the information in union regions of predicates and obtain the output $U^* = \{u_{ij}^* | i \neq j \& i, j \in N\}$. To further purify the U^* and highlight the semantics of the subject-object pair, a learnable gate is set based on the visual features of the entity pair. After that, the gated union visual features and entity pair features are sent into two projectors to align their channels. The process can be formulated as follows:

$$r_{ij}^u = \text{RelationProjector}\big(W_p\big[v_i; v_j\big] * u_{ij}^*\big) \tag{7}$$

$$r_{ij}^p = \text{PairProjector}\big(\big[v_i; v_j\big]\big) \tag{8}$$

$$r_{ij}^* = r_{ij}^u + r_{ij}^p \tag{9}$$

where r_{ij}^* is the final visual representation with the individual attributes of ith subject and jth object. Moreover, the image subprototype is required to cover general relation visual semantics across different samples. Thus, we use the lightweight head network to train a naive relation classifier to capture the general visual contents. Then, the predicate visual embedding $\{o_i\}_{i=1}^{C_r}$ can be obtained by copying the parameters (weights) $W \in \mathbb{R}^{C_r \times 2048}$ of this classifier with rich contextual semantics. Next, the image subprototype can be calculated by the following formula:

$$p_i^v = \text{MLP}^v(o_i) \tag{10}$$

where p_i^v denotes the ith image subprototype and $\text{MLP}^v(\cdot)$ denotes a two-hidden multilayer perception. The raw predicate visual embedding o_i is derived from the weights of the image-based naive classifier.

3) Subprototype Optimization: After obtaining the image and text subprototypes, the data from heterogeneous modalities needs to be aligned and compressed preliminary, which is also an important guarantee for acquiring more discriminative multimodal prototypes.

We perform intermodality contrastive learning to align the image and text subprototypes. This increases the similarity between subprototypes from different modalities that refer to the same relation predicate. This is implemented through the following optimization objective:

$$\mathcal{L}_{\text{VT}} = -\frac{1}{2C_r}\sum_{i=1}^{C_r}\left[\log\frac{\exp\Big(\Big\langle \overline{p_i^t}, \overline{p_i^v}\Big\rangle\Big)}{\sum_{j=1}^{C_r}\exp\Big(\Big\langle \overline{p_i^t}, \overline{p_j^v}\Big\rangle\Big)} + \log\frac{\exp\Big(\Big\langle \overline{p_i^v}, \overline{p_i^t}\Big\rangle\Big)}{\sum_{j=1}^{C_r}\exp\Big(\Big\langle \overline{p_i^v}, \overline{p_j^t}\Big\rangle\Big)}\right] \tag{11}$$

where $\langle\cdot,\cdot\rangle$ denotes inner product operation. $\overline{p_i^t}$ and $\overline{p_i^v}$ denote the corresponding l_2 normalized embeddings of text and image subprototypes. C_r is the number of predicate categories.

Moreover, we impose intramodality constraints to reduce the semantic overlap between subprototypes. This helps enlarge the discriminability between subprototypes of different relation predicates. The specific formulas are as follows:

$$\mathcal{L}_T = \sum_{i=0}^{C_r}\sqrt{\sum_{j=0}^{C_r}\Big\langle \overline{p_i^t}, \overline{p_j^t}\Big\rangle} \tag{12}$$

$$\mathcal{L}_V = \sum_{i=0}^{C_r}\sqrt{\sum_{j=0}^{C_r}\big\langle \overline{p_i^v}, \overline{p_j^v}\big\rangle} \tag{13}$$

where $\langle \overline{p_i^t}, \overline{p_j^t}\rangle$ and $\langle \overline{p_i^v}, \overline{p_j^v}\rangle$ represent the similarity between intramodality subprototypes. Thus, we can make each subprototype more distinctive by reducing the intramodality similarity of a pair of subprototypes.

D. Multimodal Hypergraph Learning

At this point, intermodality alignment and intramodality compression have been carried out on multimodal subprototypes. But each subprototype is still optimized independently, which overlooks relevant semantic patterns among multiple predicates. This is essentially a high-order correlation optimization problem. To conquer this problem, we devise the

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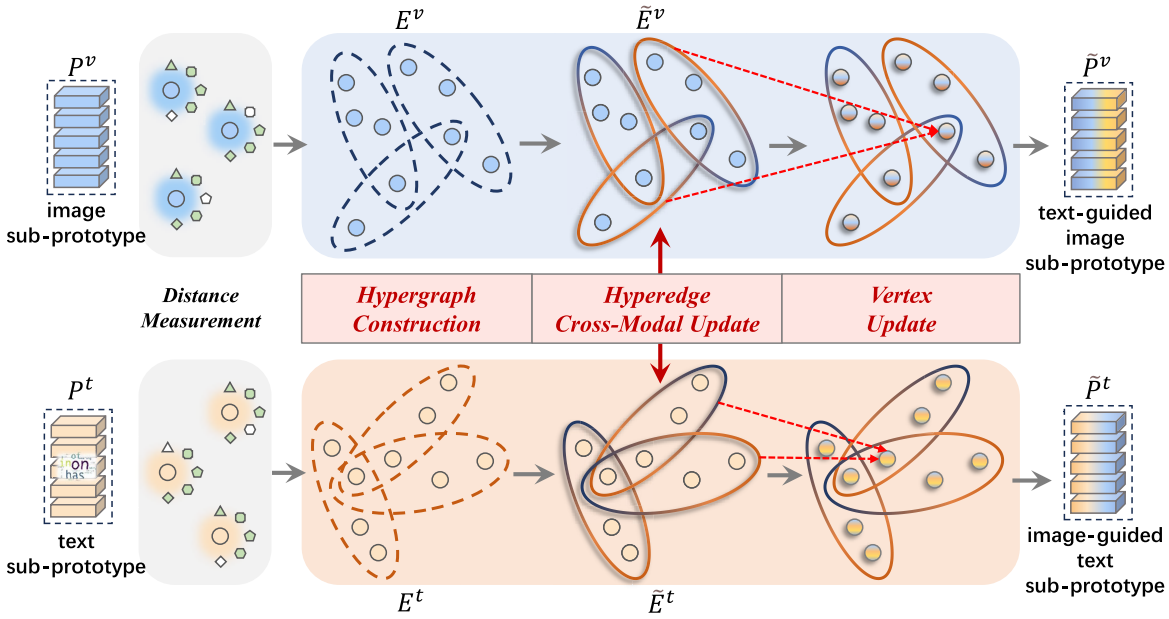


Fig. 4. Diagram of multimodal hypergraph learning. The multimodal subprototypes are treated as input vertices. The text and image intramodal hypergraphs are constructed using distance measurements. Vertex embeddings are aggregated to form the initial hyperedge features. The hyperedge-level cross-modal updating mechanism is then applied to mine relevant semantics across heterogeneous modalities. Later, the vertices are updated by receiving messages from hyperedges. Finally, the text and image subprototypes can be jointly optimized and improve the referential consistency of relation predicates.

multimodal hypergraph, which mines relevant semantic patterns among relation predicates and optimizes the referential consistency of subprototypes across heterogeneous modalities. Importantly, the design of the multimodal information transmission mechanism is crucial, as it determines whether higher order semantic associations can be effectively captured. The multimodal hypergraph learning process is illustrated in Fig. 4, and the subsequent part will provide a detailed explanation of the construction and message-updating procedures.

1) Hypergrpah Construction: To explore the associations of predicates, we first need to construct the graphical representation of the relationships among multimodal subprototypes. Formally, the intramodality relationships are described using the prototype-to-prototype text hypergraph $\mathcal{G}^t = (\mathcal{N}^t, \mathcal{E}^t)$ and image hypergraph $\mathcal{G}^v = (\mathcal{N}^v, \mathcal{E}^v)$, where $\mathcal{N}^t = \{p_i^t\}_{i=1}^{C_r}$, $\mathcal{N}^v = \{p_i^v\}_{i=1}^{C_r}$, $\mathcal{E}^t = \{\mathcal{E}_i^t\}_{i=1}^{M_t}$, and $\mathcal{E}^v = \{\mathcal{E}_i^v\}_{i=1}^{M_v}$ are the collections of vertices and hyperedges. Here, the text and image subprototypes processed by the optimization strategy obtained in the last subsection are taken as the input vertex embeddings of multimodal hypergraphs. In addition, the topological structures of hypergraphs can be summarized by incidence matrices $\mathbf{H}^t \in \mathbb{R}^{C_r \times M_t}$ and $\mathbf{H}^v \in \mathbb{R}^{C_r \times M_v}$

$$\mathbf{H}_{ij}^t = \begin{cases} 1, & \text{if } p_i^t \in \mathcal{E}_j^t \\ 0, & \text{if } p_i^t \notin \mathcal{E}_j^t \end{cases} \tag{14}$$

$$\mathbf{H}_{ij}^v = \begin{cases} 1, & \text{if } p_i^v \in \mathcal{E}_j^v \\ 0, & \text{if } p_i^v \notin \mathcal{E}_j^v. \end{cases} \tag{15}$$

It can be seen from this that these hyperedges are the high-order connections of subsets of vertices, and hypergraph construction is the process of searching vertex clusters. To capture the associations among vertices and connect subprototypes enjoying relevant semantics, we measure the cosine distance of vertex embedding pairs and keep the nearest K neighbors for each central vertex to construct the corresponding initial hyperedges, which can be formulated as follows:

$$\mathcal{E}_i^t = p_i^t \cup \left\{p_n^t, p_{n+1}^t, \ldots, p_{n+K-1}^t\right\} \tag{16}$$

$$\mathcal{E}_i^v = p_i^v \cup \left\{p_m^v, p_{m+1}^v, \ldots, p_{m+K-1}^v\right\}. \tag{17}$$

When each hyperedge is determined, the incidence matrices $\mathbf{H}^t$ and $\mathbf{H}^v$ can also be solved, and the construction of multimodal hypergraphs is finished.

2) Hyperedge Cross-Modal Update: Most existing works about hyperedges focus on the message-passing mechanism within the single hypergraph to iteratively update representations or deep features. Compared with that, to enable information interaction between multimodal multiple hypergraphs, a hyperedge-based cross-modal updating approach is devised to achieve complementary guidance of context across different modalities. This is also the key that multimodal subprototypes with relevant semantics can be jointly optimized. The flowchart of the hyperedge-level cross-modal updating mechanism is shown in Fig. 5.

Given the vertices $\{p_k^t\}_{k=1}^K$ or $\{p_k^v\}_{k=1}^K$ contained on the ith hyperedge $\mathcal{E}_i^t$ or $\mathcal{E}_i^v$, we first measure the contribution of each vertex to the hyperedge and calculate the weighted sum as the initial hyperedge feature, which can be formulated as follows:

$$\alpha_{ik} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}_1^\top \mathbf{W}_1^t p_k^t\right)\right)}{\sum_{p_q^t \in \mathcal{E}_i^t} \exp\left(\text{LeakyReLU}\left(\mathbf{a}_1^\top \mathbf{W}_1^t p_q^t\right)\right)} \tag{18}$$

$$e_i^t = \sum_{p_k^t \in \mathcal{E}_i^t} \alpha_{ik} \mathbf{W}_1^t p_k^t \tag{19}$$

where $\mathbf{W}_1^t$ and $\mathbf{a}_1$ are the trainable parameters for dimension transformation. The activation function leaky rectified linear unit (LeakyReLU) is added to optimize training. α_{ik} denotes

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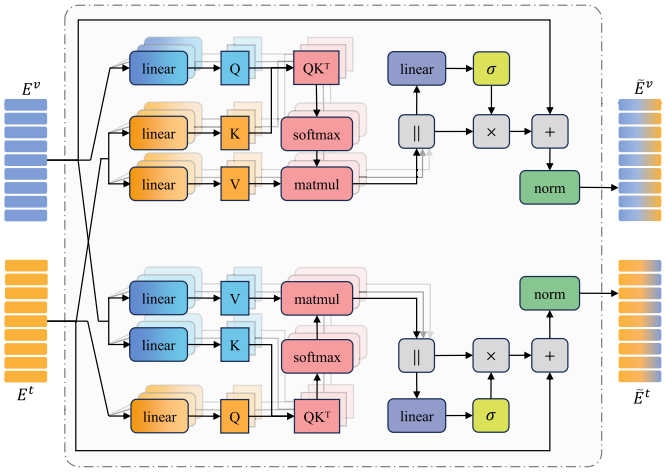


Fig. 5. Illustration of the cross-modal updating mechanism on the hyperedge. The embeddings of multimodal hyperedges are interacted by cross-modal attention and the output results are cleaned via the adaptive gate function. In this way, the multimodal messages can flow in a cross-hypergraph manner and achieve high-order aggregation across different modalities.

the contribution coefficient of the kth vertex on the hyperedge $\mathcal{E}_i^t$. e_i^t is the initial text hyperedge feature obtained through weighted summation. Also, the initial image hyperedge feature e_i^v can be obtained via the similar formulas.

To enable effective multimodal interaction, the cross-modal updating mechanism on hyperedges needs to allow the exchange of complementary information between any two hyperedges, regardless of their topological positions in the multimodal hypergraphs. This inspires us to develop the attention mechanism [77] on multimodal hyperedges to model distance-free dependencies. Specifically, we define E^t/E^v as the batch of initial hyperedge features e_i^t/e_i^v

$$E^t = \left[e_1^t, e_2^t, \ldots, e_{M_t}^t\right]^\top \tag{20}$$

$$E^v = \left[e_1^v, e_2^v, \ldots, e_{M_v}^v\right]^\top \tag{21}$$

where $[\cdot]^\top$ represents the batch packaging operation. M_t and M_v are the numbers of hyperedges. We first get the required multimodal variables including query Q_h^t/Q_h^v, key K_h^t/K_h^v, and value V_h^t/V_h^v through the corresponding linear transformation

$$\begin{cases} Q_h^t = E^t W_h^{Q_t}, & K_h^t = E^t W_h^{K_t}, \quad V_h^t = E^t W_h^{V_t} \\ Q_h^v = E^v W_h^{Q_v}, & K_h^v = E^v W_h^{K_v}, \quad V_h^v = E^v W_h^{V_v} \end{cases} \tag{22}$$

where $W_h^{(\cdot)}$ is the parameters of linear transformation. $h \in \{1, 2, \ldots, N_h\}$ and N_h is the number of attention heads.

Then, we calculate the cross-attention weights and obtain the weighted results of the hth single head pair, including the text-to-image attention-weighted output $(\text{head})_h^{t-v}$ and the image-to-text attention-weighted output $(\text{head})_h^{v-t}$

$$\begin{aligned}(\text{head})_h^{t-v} &= \text{Atten}\left(Q_h^t, K_h^v, V_h^v\right) \\ &= \left[\text{Softmax}\left(\left(Q_h^t\left(K_h^v\right)^\top\right)\Big/\sqrt{d_k}\right)\right]V_h^v \end{aligned} \tag{23}$$

$$\begin{aligned}(\text{head})_h^{v-t} &= \text{Atten}\left(Q_h^v, K_h^t, V_h^t\right) \\ &= \left[\text{Softmax}\left(\left(Q_h^v\left(K_h^t\right)^\top\right)\Big/\sqrt{d_k}\right)\right]V_h^t \end{aligned} \tag{24}$$

where d_k is a commonly used scaling factor. In this way, the image modality uses the query Q_h^v to extract useful contents in the V_h^t that is derived from the text modality, and vice versa.

Next, we combine outputs of all single attention heads to obtain the batch of cross-modal updated hyperedge features E^{t-v}/E^{v-t}, which can be calculated as follows:

$$E^{t-v} = \left[(\text{head})_1^{t-v}; (\text{head})_2^{t-v}; \ldots; (\text{head})_{N_h}^{t-v}\right] W^{O_1} \tag{25}$$

$$E^{v-t} = \left[(\text{head})_1^{v-t}; (\text{head})_2^{v-t}; \ldots; (\text{head})_{N_h}^{v-t}\right] W^{O_2} \tag{26}$$

where $[\cdot;\cdot]$ denotes the concatenation operation. W^{O_1} and W^{O_2} represent linear layers used to integrate information from multiple attention heads.

At last, we set up the gate function to further filter the noise after cross-modal interaction, and add the purified result with the identity mapping value to obtain the final updated hyperedge features. The formulas are expressed as follows:

$$G^{t-v} = \sigma\left(W^{G_1}\left[E^t; E^{t-v}\right]\right), \quad G^{v-t} = \sigma\left(W^{G_2}\left[E^v; E^{v-t}\right]\right) \tag{27}$$

$$\widetilde{E}^t = \text{LayerNorm}\left(E^t + G^{t-v} \odot E^{t-v}\right) \tag{28}$$

$$\widetilde{E}^v = \text{LayerNorm}\left(E^v + G^{v-t} \odot E^{v-t}\right) \tag{29}$$

where W^{G_1} and W^{G_2} denote the fully connected layers used to adjust the dimensions. $[\cdot;\cdot]$ represents the concatenation operation. $\sigma(\cdot)$ denotes sigmoid activation function. $\odot$ is the elementwise product operation. Layer normalization is adopted to make the training more stable. During the cross-modal interaction shown in (28) and (29), the gates G^{t-v} and G^{v-t} can weaken the dispersion of the origin critical semantics and help focus on complementary contextual information of the corresponding modality.

At this point, the barriers between text and image modalities are broken down. The cross-modal updated hyperedge features $\widetilde{E}^t$ and $\widetilde{E}^v$ not only aggregate the information of the vertices in their respective modality but also supplement the semantic content of the other modality.

3) Vertex Update: In this step, the vertex update process is used to aggregate multimodal context-rich hyperedge features and propagate them to the vertices inside the text and image hypergraphs. Formally, the set of hyperedges connected to vertex p_k^t or p_k^v is $\mathcal{D}_k^t = \{\mathcal{E}_1^t, \mathcal{E}_2^t, \ldots, \mathcal{E}_{D_k}^t\}$ or $\mathcal{D}_k^v = \{\mathcal{E}_1^v, \mathcal{E}_2^v, \ldots, \mathcal{E}_{D_k}^v\}$, then the text vertex update process is

$$\widetilde{E}^t = \left[\widetilde{e}_1^t, \widetilde{e}_2^t, \ldots, \widetilde{e}_{M_t}^t\right]^\top \tag{30}$$

$$\widetilde{E}^v = \left[\widetilde{e}_1^v, \widetilde{e}_2^v, \ldots, \widetilde{e}_{M_v}^v\right]^\top \tag{31}$$

$$\beta_{ki} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}_2^\top \mathbf{W}_2^t \widetilde{e}_i^t\right)\right)}{\sum_{\mathcal{E}_q^t \in \mathcal{D}_k^t} \exp\left(\text{LeakyReLU}\left(\mathbf{a}_2^\top \mathbf{W}_2^t \widetilde{e}_q^t\right)\right)} \tag{32}$$

$$\widetilde{p}_k^t = \sum_{\mathcal{E}_i^t \in \mathcal{D}_k^t} \beta_{ki} \mathbf{W}_2^t \widetilde{e}_i^t \tag{33}$$

where $\mathbf{W}_2^t$ and $\mathbf{a}_2$ are the learnable parameters. β_{ki} denotes the contribution coefficient of the ith hyperedge in $\mathcal{D}_k^t$. $\widetilde{e}_i^t$ sampled from the batch $\widetilde{E}^t$ is the cross-modal updated feature of the specific hyperedge named $\mathcal{E}_i^t$. $\widetilde{p}_k^t$ is the updated text vertex embedding. In addition, we calculate the updated image vertex embedding $\widetilde{p}_k^v$ in a similar way following (33). In this way, originally single-modality subprototype vertices can

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achieve deep cross-modal information fusion after aggregating messages from multiple heterogeneous hypergraphs. Thus, after the intrahypergraph hyperedge-to-vertex update, we can obtain the text-guided image subprototypes $\widetilde{P}^v = \{\widetilde{p}_k^v\}_{k=1}^{C_r}$ and image-guided text subprototypes $\widetilde{P}^t = \{\widetilde{p}_k^t\}_{k=1}^{C_r}$. C_r is the number of relation predicate categories.

As a result, beyond the intrahypergraph update and the intramodality aggregation, the proposed multimodal hypergraph benefits from a high-order cross-modal interaction mechanism, thereby helping to efficiently mine relevant semantics among image and text subprototypes.

E. Multimodal Calibration and Matching Classification

To make the representation-prototype matching process more modality-balance and improve the reliability of classification, the multimodal calibration strategy is devised based on the multimodal subprototypes with sufficient modality knowledge and semantic integrity.

The text subprototype contains more accurate semantics and we calculate the text subprototype-based (TSP-based) matching logits S^t using the following formula:

$$S^t = \left\{ q_k^t \middle| q_k^t = \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^t} \right\rangle \right\}_{k=1}^{C_r}. \tag{34}$$

The image subprototype can carry more complex and plentiful content and we calculate the image subprototype-based (ISP-based) matching logits S^v as follows:

$$S^v = \left\{ q_k^v \middle| q_k^v = \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^v} \right\rangle \right\}_{k=1}^{C_r}. \tag{35}$$

We optimize the matching logits S^v and S^t at the same time to calibrate the distribution of the matching classification. For both ISP-based and TSP-based matching, we need to pull the predicate representations and predicate prototypes belonging to the same category closer and push them away from the representations and prototypes of the other categories. In this way, through competitive equilibrium learning of multimodal prototypes, a more reasonable metric space is obtained to better measure the multimodal components of relation samples. The modal-competitive objective functions are as follows:

$$\mathcal{L}_{R_t} = -\frac{1}{N_r} \sum_{ij} \log \frac{\exp\left(\left\langle \overline{r_{ij}}, \overline{\widetilde{p}_l^t} \right\rangle \Big/ \tau_t \right)}{\sum_{k=1}^{C_r} \exp\left(\left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^t} \right\rangle \Big/ \tau_t \right)} \tag{36}$$

$$\mathcal{L}_{R_v} = -\frac{1}{N_r} \sum_{ij} \log \frac{\exp\left(\left\langle \overline{r_{ij}}, \overline{\widetilde{p}_l^v} \right\rangle \Big/ \tau_v \right)}{\sum_{k=1}^{C_r} \exp\left(\left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^v} \right\rangle \Big/ \tau_v \right)} \tag{37}$$

where N_r denotes the number of relation samples and τ_t/τ_v are temperature hyperparameters. $\widetilde{p}_l^t$ and $\widetilde{p}_l^v$ are the multimodal subprototypes corresponding to the ground-truth category.

In the inference stage, we devise a refinement calibration strategy to calibrate the predictions further when matching relation samples with multimodal subprototypes. Specifically, we try to combine the advantages of the two matching logits to perform sample-level refinement calibration. We first use softmax function δ to get the TSP-based and ISP-based matching scores G^t and G^v

$$G^t = \left\{ g_k^t \right\}_{k=1}^{C_r} = \delta\left(q_k^t \middle| \left\{ \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^t} \right\rangle \right\}_{k=1}^{C_r} \right) \tag{38}$$

$$G^v = \left\{ g_k^v \right\}_{k=1}^{C_r} = \delta\left(q_k^v \middle| \left\{ \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^v} \right\rangle \right\}_{k=1}^{C_r} \right) \tag{39}$$

where r_{ij} denotes a relation sample to be predicted, i.e., the predicate representation corresponding to the ith subject and the jth object. We devise an adaptive strategy to calculate the confidence of the matching scores for both

$$s_{\max}^t = \max\left(\left\{ g_k^t \right\}_{k=1}^{C_r} \right), \quad s_{\max}^v = \max\left(\left\{ g_k^v \right\}_{k=1}^{C_r} \right) \tag{40}$$

$$\theta_t = -\sum_k g_k^t \log\left(g_k^t\right), \quad \theta_v = -\sum_k g_k^v \log\left(g_v^t\right) \tag{41}$$

$$\lambda^t = s_{\max}^t \exp\left(-2{\theta_t}^2\right), \quad \lambda^v = s_{\max}^v \exp\left(-2{\theta_v}^2\right) \tag{42}$$

where λ^t/λ^v are the confidence values coupled by the maximum score $s_{\max}^t/s_{\max}^v$ and entropy θ_t/θ_v. This confidence measures the prediction quality and distribution certainty of these matching scores. Then, we choose the scores with higher confidence to predict the predicate category for the sample r_{ij}

$$c_{\text{res}} = \begin{cases} \arg\max_k \left(\left\{ g_k^t \right\}_{k=1}^{C_r} \right), & \text{if } \lambda^t > \lambda^v \\ \arg\max_k \left(\left\{ g_k^v \right\}_{k=1}^{C_r} \right), & \text{if } \lambda^t \le \lambda^v \end{cases} \tag{43}$$

where c_{res} is the predicate class index of the final result.

F. Loss Function

Considering the optimization objectives of each part, the overall training loss of HMPNet can be defined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{VRC}} + \mathcal{L}_{\text{VT}} + \mathcal{L}_T + \mathcal{L}_V + \mathcal{L}_{R_t} + \mathcal{L}_{R_v} \tag{44}$$

where $\mathcal{L}_{\text{VRC}}$ represents the cross-entropy loss for the simple visual relation classifier described in Section III-C2.

IV. Experiments

In this section, we describe the experimental setup, including data details, model configurations, and metrics. We conduct experiments on several remote sensing scene graph tasks and analyze the results. Ablation studies are performed to show the effectiveness of each model component, and visualization results are provided to qualitatively assess performance.

A. Datasets Description

We construct a novel large-scale remote sensing scene understanding dataset named RSSUD, which is a relation recognition dataset developed for promoting remote sensing scene understanding research. We provide the details on the data collection and dataset statistics below.

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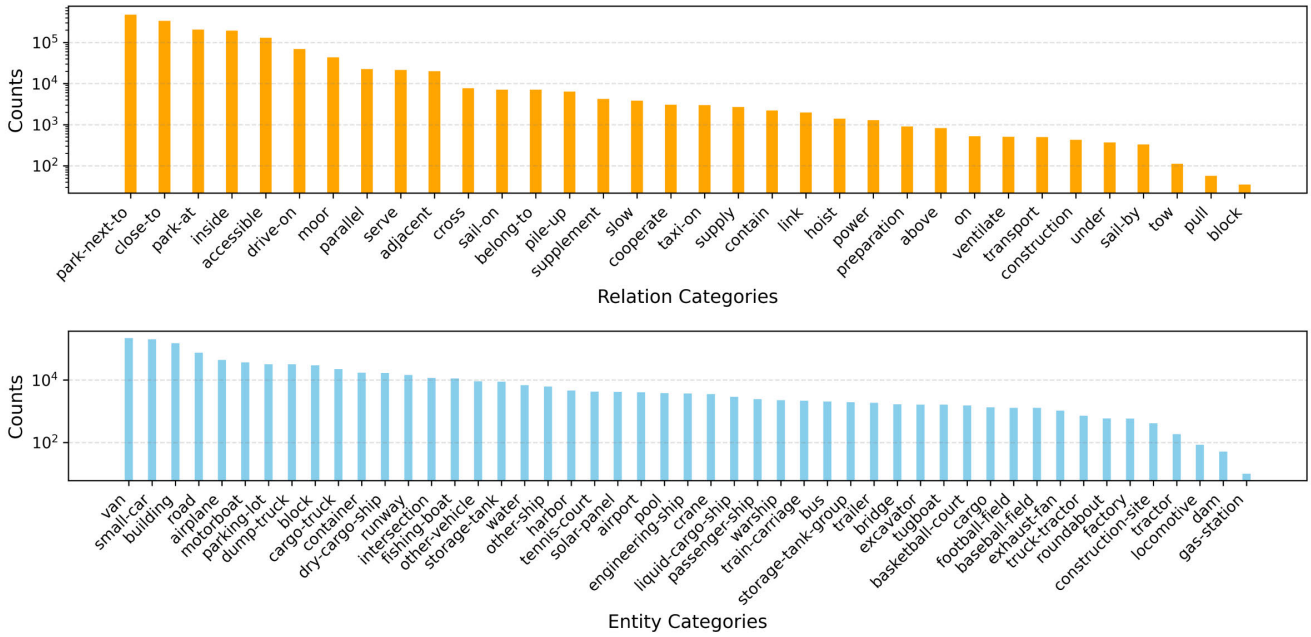


Fig. 6. Statistics of relation categories and entity categories on the RSSUD dataset. For better presentation, the results are plotted on a logarithmic scale.

1) Data Collection: We choose the million-level object detection dataset FAIR1M [5] with fine-grained entity labels as our image source. We perform three rounds of annotation via crowd-sourcing services. In the first round, these images with too few objects or no meaningful relations are filtered out. In the second round, to fully explore the scene information, we expand the entity categories to 49, including van, small car, building, road, airplane, block, parking lot, motorboat, dump truck, cargo truck, dry cargo ship, runway, container, water, intersection, fishing boat, other vehicle, storage tank, airport, other ship, harbor, engineering ship, tennis court, pool, solar panel, liquid cargo ship, crane, bus, passenger ship, warship, storage tank group, excavator, bridge, tugboat, basketball court, trailer, train carriage, football field, cargo, baseball field, exhaust fan, truck tractor, factory, roundabout, construction site, tractor, dam, locomotive, and gas-station. The positions of all entities are annotated with oriented bounding boxes. After that, the third round focuses on the annotation of the relations between entities. There are 34 meaningful predicate categories, including 11 categories of spatial relations, i.e., park next to, close to, park at, accessible, parallel, adjacent, inside, cross, above, under, on, and 23 categories of semantic relations, i.e., drive on, moor, serve, sail on, belong to, pile up, supplement, supply, slow, contain, taxi on, cooperate, power, link, preparation, hoist, ventilate, transport, construction, sail by, tow, block, and pull. Finally, multiple inspections are carried out under the guidance of experts.

2) Dataset Statistics: As a result, the RSSUD dataset contains large-scale 30 494 images with a resolution of 0.3–0.8 m. The RSSUD dataset contains 1 006 802 entity targets annotated by oriented bounding boxes and there are 49 various entity categories. The total number of relations between entity pairs is 1 583 137 across 34 different predicate categories. The quantity distribution of each relation and entity category is shown in Fig. 6. It can be seen that RSSUD is a large-scale challenging dataset for remote sensing scene understanding, which involves the high-level relation judgments between entities with varying categories. The strong diversity and complexity are more conducive to testing the performance and robustness of the model. Furthermore, the training set and test set are divided in a ratio of 3:1, resulting in 22 989 and 7505 images, respectively. All models are trained on the training set and the reported results are achieved on the test set.

B. Implementation Details

In all experiments, we employ a pretrained Rotated Faster R-CNN [78] with the backbone ResNet-50 [79] as the oriented detector to predict bounding boxes and categories of potential targets, and the input size is set to 800 × 800. The parameters of the oriented detector are frozen when training the relation predictors for the SGG task. We choose the Adam [80] as our optimizer and all experiments are implemented on a single A-800 GPU with Pytorch [81] framework. We use the cosine annealing scheduling strategy [82] and set the initial learning rate to 0.0001. The batch size of our model is 4 and the total training epochs are 24.

C. Evaluation and Tasks

1) Tasks: We evaluate the proposed model on three remote sensing SGG subtasks, i.e., predicate classification (PredCls), scene graph classification (SGCls), and scene graph detection (SGDet). To be specific, when performing the PredCls subtask, the ground-truth bounding boxes and labels of entities are given, and the model predicts the relation labels. For the SGCls subtask, the ground-truth bounding boxes of entities are given, and the model predicts the entity labels and relation labels. For

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TABLE I

OVERALL PERFORMANCE COMPARISONS WITH DIFFERENT METHODS ON PREDCLS, SGCLS, AND SGDET TASKS. THE REMOTE SENSING SGG RESULTS OF R@20/50/100/500, MR@20/50/100/500, AND H@20/50/100/500 ARE REPORTED. "BERT" AND "CT" DENOTE THAT THE TEXT EMBEDDING IS OBTAINED BY THE BERT METHOD AND CLIP-TEXT ENCODER, RESPECTIVELY

Recall@K												
Method	PredCls				SGCls				SGDet			
	R@20	R@50	R@100	R@500	R@20	R@50	R@100	R@500	R@20	R@50	R@100	R@500
IMP [49]	59.4	75.5	84.7	93.6	53.4	66.4	73.5	79.6	26.4	32.9	37.6	44.1
Motifs [41]	61.8	78.2	87.6	96.5	53.5	66.6	73.3	79.5	27.3	33.9	38.0	44.3
RTN [83]	62.4	78.9	88.3	97.1	53.6	66.8	73.8	80.3	27.7	34.0	38.1	44.4
VCTree [42]	62.1	78.5	87.9	96.8	53.5	66.5	73.3	79.6	27.6	34.1	38.1	44.5
GPS-Net [43]	62.3	78.8	88.1	96.7	49.8	62.2	69.1	76.0	27.6	34.1	38.2	44.5
G-RCNN [50]	61.8	78.3	87.6	96.6	49.5	62.0	68.9	75.8	26.9	33.7	38.0	44.5
SQUAT [84]	61.5	78.1	87.4	96.6	49.0	61.4	68.3	75.5	26.6	33.2	37.8	44.5
PE-Net [48]	61.8	78.2	87.6	96.6	52.5	65.1	71.8	77.8	27.8	34.1	38.2	44.5
HiKER [85]	62.1	78.6	88.1	97.0	52.6	65.4	72.0	79.2	27.6	34.0	38.2	44.2
HMPNet	62.7	**79.2**	**88.6**	97.3	53.8	66.9	73.9	80.4	**27.8**	34.1	38.3	44.3
HMPNet+BERT [86]	62.7	79.2	88.6	97.3	54.0	**67.2**	**74.2**	80.7	27.8	**34.2**	**38.4**	**44.5**
HMPNet+CT [75]	**62.8**	79.2	88.6	**97.4**	**54.0**	67.1	74.1	**80.7**	27.7	34.0	38.2	44.2

Mean Recall@K												
Method	PredCls				SGCls				SGDet			
	mR@20	mR@50	mR@100	mR@500	mR@20	mR@50	mR@100	mR@500	mR@20	mR@50	mR@100	mR@500
IMP [49]	30.5	45.0	55.8	70.5	29.0	42.1	50.6	62.0	11.1	15.7	19.8	26.7
Motifs [41]	38.5	53.8	64.3	81.1	30.5	42.8	50.8	62.2	11.6	15.8	19.8	26.1
RTN [83]	44.1	60.3	72.6	90.1	30.5	43.2	52.0	63.3	12.3	16.9	20.9	27.0
VCTree [42]	40.5	56.4	67.4	83.8	29.6	42.1	51.0	62.5	12.1	16.2	20.4	27.2
GPS-Net [43]	41.1	57.3	70.8	89.8	26.0	36.3	43.6	54.7	12.2	16.6	21.1	27.8
G-RCNN [50]	36.5	52.6	64.8	82.8	24.5	35.4	42.4	53.6	11.4	16.1	20.4	27.9
SQUAT [84]	40.0	56.4	70.2	88.5	25.1	35.5	42.8	53.1	12.0	16.5	21.0	27.8
PE-Net [48]	45.6	60.7	71.4	88.8	28.7	40.6	48.4	58.6	11.9	16.0	19.9	26.9
HiKER [85]	46.0	61.3	73.2	90.2	30.1	40.6	47.6	58.4	11.8	16.1	20.1	26.8
HMPNet	**48.9**	**64.7**	**76.1**	**92.1**	31.9	44.9	54.0	64.7	13.0	**17.7**	**21.9**	**28.1**
HMPNet+BERT [86]	48.3	64.6	75.5	91.8	**32.8**	**45.5**	**55.1**	**65.5**	**13.1**	17.4	21.8	27.9
HMPNet+CT [75]	47.7	64.1	75.7	92.1	32.1	45.3	54.4	65.4	12.8	17.3	21.2	27.4

Harmonic Mean@K												
Method	PredCls				SGCls				SGDet			
	H@20	H@50	H@100	H@500	H@20	H@50	H@100	H@500	H@20	H@50	H@100	H@500
IMP [49]	40.3	56.4	67.3	80.4	37.6	51.5	59.9	69.7	15.6	21.3	25.9	33.3
Motifs [41]	47.4	63.7	74.2	88.1	38.9	52.1	60.0	69.8	16.3	21.6	26.0	32.8
RTN [83]	51.7	68.4	79.7	93.5	38.9	52.5	61.0	70.8	17.0	22.6	27.0	33.6
VCTree [42]	49.0	65.6	76.3	89.8	38.1	51.6	60.1	70.0	16.8	22.0	26.6	33.8
GPS-Net [43]	49.5	66.4	78.5	93.1	34.2	45.8	53.5	63.6	16.9	22.3	27.2	34.2
G-RCNN [50]	45.9	62.9	74.5	89.2	32.8	45.1	52.5	62.8	16.0	21.8	26.5	34.3
SQUAT [84]	48.5	65.5	77.9	92.4	33.2	45.0	52.6	62.3	16.5	22.0	27.0	34.2
PE-Net [48]	52.5	68.3	78.7	92.5	37.1	50.0	57.8	66.8	16.7	21.8	26.2	33.5
HiKER [85]	52.8	68.9	79.9	93.5	38.3	50.1	57.3	67.2	16.5	21.9	26.3	33.4
HMPNet	**54.9**	**71.2**	**81.9**	94.6	40.1	53.7	62.4	71.7	17.7	**23.3**	**27.9**	**34.4**
HMPNet+BERT [86]	54.6	71.1	81.5	94.5	**40.8**	**54.3**	**63.2**	**72.3**	**17.8**	23.1	27.8	34.3
HMPNet+CT [75]	54.2	70.9	81.6	**94.7**	40.3	54.1	62.7	72.2	17.5	22.9	27.3	33.8

the SGDet subtask, the model is required to predict bounding boxes, labels of entities, and their relation predicates.

2) Metrics: We quantitatively evaluate the performance of the model using three evaluation metrics, including recall@K (R@K), mean recall@K (mR@K), and harmonic mean@K (H@K). The R@K is a common SGG metric that performs calculations on the top K predictions of all relation samples, which is easily dominated by head predicate categories with more samples. Then, mR@K [51] is devised to take into account all relation categories and is obtained by a category-averaged R@K. To comprehensively measure the overall performance, we also adopt the harmonic mean to combine the above indicators and pay better attention to the smaller value.

D. Comparisons With State-of-the-Art (SOTA) Methods

To evaluate the performance of the proposed model on the remote sensing relation comprehension, we compare it with some popular SOTA SGG models under different subtask settings, and the results are reported in Table I.

1) Main Results: In general, the proposed model HMPNet achieves 64.7%/76.1% on mR@50/100 and 71.2%/81.9% on H@50/100 for the pure relation recognition task (i.e., PredCls), which is the best accuracy among these

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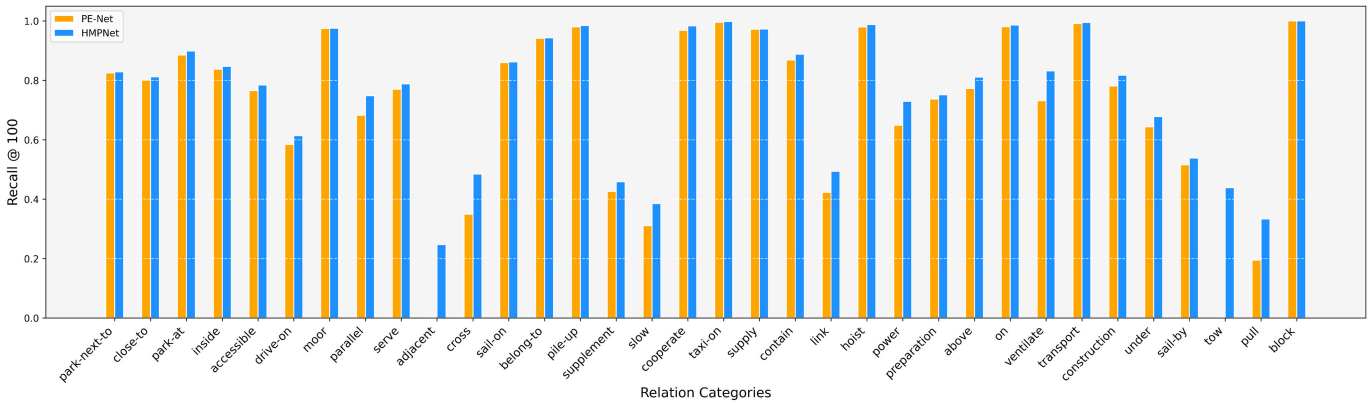


Fig. 7. Performance comparison between single-modal prototype-based PE-Net and the proposed multimodal prototype-based HMPNet. Judging from the Recall@100 of each relation category on the PredCls task, HMPNet can boost the performance for some hard categories after obtaining more discriminative prototypes, and it improves the accuracy of overall classes by reasoning relevant semantic patterns.

reported methods under the same experimental settings. For more challenging task settings, our model can achieve 53.7%/62.4% on H@50/100 of SGCls task and 23.3%/27.9% on H@50/100 of SGDet task, demonstrating that our model can distinguish different categories of relation predicates more accurately and produce remote sensing scene graph more effectively. Furthermore, HMPNet showcases significant improvements of more than 2.0%, 1.4%, and 0.7% in terms of H@100 on the PredCls, SGCls, and SGDet tasks, indicating the stronger superiority of the hypergraph-guided framework. The proposed HMPNet overcomes the problem of modal imbalance and effectively leverages the associated semantic patterns among predicates. Thus, it achieves the expected progress across the relevant metrics. These findings reveal that the proposed multimodal-prototype-based paradigm is valuable for understanding complex remote sensing scenes.

2) Quality of Relation Prediction: The prediction quality and reliability of the model are also important considerations in practical applications. The predictions with the higher confidence scores correspond to the top-ranked testing samples, and the correct proportion of the leading samples with higher scores can reflect the reliability of the model to some extent. Thus, we report the accuracy in terms of R@20, mR@20, and H@20 to examine the aforementioned aspect. As shown in Table I, the proposed HMPNet achieves an impressive mR@20 of 48.9% on the PredCls subtask, surpassing other methods by a large margin (more than 2.9%). This verifies the advantage of our model on the quality of the relation prediction is more obvious. Note that the G-RCNN [50] adopts a relation proposal network to select high-quality subject–predicate–object triplets. Nevertheless, HMPNet outperforms it by 9.0% on H@20 (45.9% versus 54.9%), indicating that the refinement and screening of relation triplets in the complex remote sensing scene is more challenging. This also shows that improving the discriminability of predicate prototypes is effective in improving the quality of relation prediction.

3) Multimodal Prototypes Versus Single-Modal Prototype: Compared with the baseline model PE-Net [48] that encodes the single-modal prototype only by the text embedding, the proposed HMPNet lifts the R@20 from 61.8% to 62.7% on the PredCls task because the modality-balance representation-prototype matching process is achieved and the mining of relevant semantic patterns of predicates is conducted by the multimodal message updating via hypergraphs. Furthermore, HMPNet increases the mR@20 by 3.3% on PredCls because the referential content of our prototype obtained based on aligned image and text modalities is clearer and more discriminative. To better showcase the effectiveness of the multimodal prototype, the Recall@100 per relation category on the PredCls task is reported in Fig. 7. From the class-specific comparative analysis, HMPNet can boost the performance of some hard relation categories, e.g., from 0.27% to 24.7% for "adjacent" and from 0.0% to 43.9% for "tow." This indicates the proposed multimodal prototype with sufficient complementary semantics is more accurate than the text-only modeling approach. Moreover, compared with the PE-Net that uses the independent prototypes, HMPNet can further lift the accuracy of relation categories with relevant contents, such as "drive on", "sail on," and "sail by." For relations with some similar metaphorical meanings (e.g., "pull" and "link", "on," and "above"), there are also improvements because the hypergraph reasoning helps capture intersection information among them. Hence, multimodal prototypes are useful and can collaboratively facilitate remote sensing scene understanding.

4) Comparative Analysis: Some classic SGG models for scene understanding, such as IMP [49], Motifs [41], and VCTree [42], they can give acceptable values of R@20/50/100/500 on three tasks of remote sensing images and HMPNet outperforms them in all metrics. Compared with VCTree [42] that learns visual contexts via dynamic tree structures, our HMPNet lifts the mR@50/100 on SGCls task by more than 2.8%/3.0% due to the improvements of each relation category via the predicate-specific prototype-based approach. Compared with the modern model SQUAT [84] that adopts stacked high-cost global attention modules to refine entities and pairs, HMPNet lifts the R@20 value from 26.6% to 27.8% and mR@20 value from 12.0% to 13.0% on the demanding SGDet task due to the more reliable multimodal prototypes with the more efficient hypergraph guidance.

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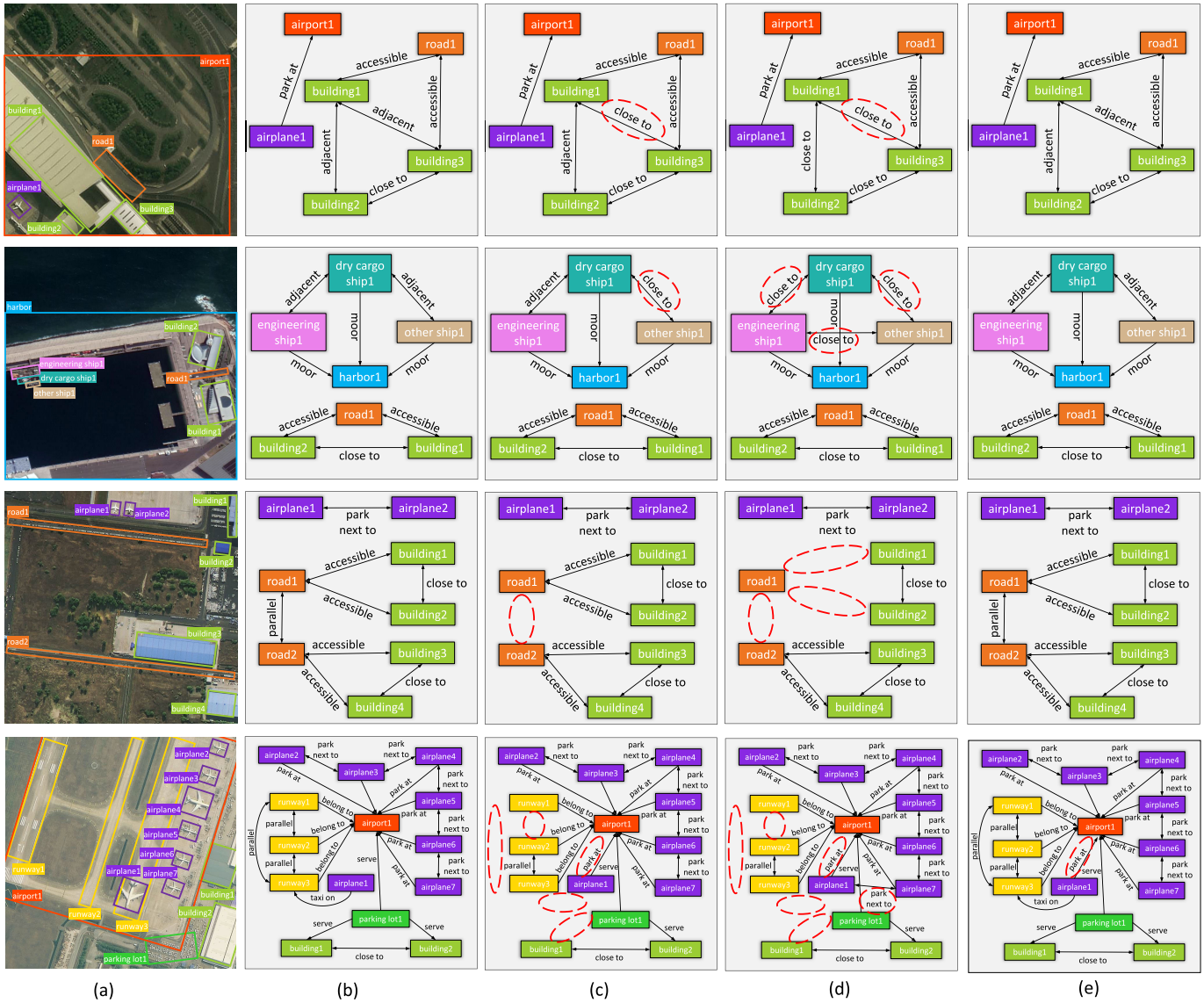


Fig. 8. Visualization comparisons of different models. (a) Oriented bounding boxes of entities. (b) Ground-truth remote sensing scene graph. (c) and (d) Predicted results of SQUAT and PE-Net. (e) Output scene graph of the proposed HMPNet. The red dashed ellipses show the relation predictions that are wrong or missed. Our HMPNet can generate more accurate and more informative remote sensing scene graphs.

Compared with the recent model HiKER [85] that utilizes knowledge graph to purify the coarse results to obtain more detailed predictions, our HMPNet brings the H@100 improvements of 2.0% and 1.6% on the PredCls and SGDet tasks, indicating that prototypes updated by high-order hypergraphs have more superiority in determining the relations between oriented remote sensing targets. Hence, all results in Table I show that our HMPNet has strong comprehensive performance and can become a new potential baseline model for remote sensing SGG tasks.

5) Exploration for Different Text Embedding Methods: We investigate the performance with different methods of text embedding. In Table I, we test HMPNet using more advanced text encoding techniques, i.e., BERT [86] and CLIP-text [75] encoder. The overall performance of HMPNet + BERT is better than that of HMPNet + CT. This indicates that the text-specific BERT model is more fine-grained for word encoding and more suitable for the proposed framework than CLIP pretrained using image-level caption texts. Furthermore, for the PredCls task, the performance of models using different text encoding methods is similar. This may be because the entity and predicate labels are short texts and the discrimination gap is not large when given the clear entity labels. For the SGCls task involving noise of entity labels, compared to the vanilla HMPNet, the HMPNet + BERT model improves the H@100 by 0.8% because the text embeddings derived from the advanced method enjoy greater potential to work with image modality to obtain more accurate relation results. For the SGDet task, the accuracy difference is not significant because the detected entity bounding boxes and labels cause both the image modality and the text modality to contain more noise, making the multimodal learning process more challenging.

6) Visualization Results: We provide some visualization results to compare the performances of different models qualitatively. As shown in Fig. 8, the SQUAT model misses

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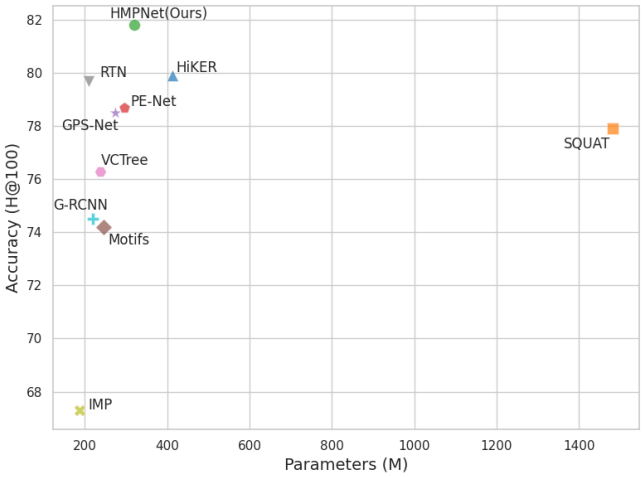


Fig. 9. Comparison of parameter quantity and performance on the PredCls task. The comprehensive performance of the proposed HMPNet significantly surpasses that of other popular models, achieving the best tradeoff between parameters and performance.

some critical relations in the dense scenario even though it uses cascade attention modules to process features of entities and relations. The PE-Net model sometimes confuses some relation categories (e.g., "close to" and "adjacent") due to insufficient information from text-only predicate prototypes. In various remote sensing scenes such as the airport and harbor, the proposed hypergraph-guided multimodal prototypes can help the model gain stronger discriminability and give more accurate results of relation recognition. Besides, our HMPNet can generate more complete scene graphs and better facilitate the understanding of remote sensing scenes.

7) Comparison of Model Complexity: We compare the model complexity to further verify the practicality of the proposed framework. The parameters of different models are illustrated in Fig. 9. It can be seen that the hypergraph-based HMPNet enjoys the sparser structure and has much fewer parameters than the SQUAT model which is based on stacked dense global attention blocks. Compared with the text-based PE-Net, our HMPNet with multimodal prototypes uses a similar number of parameters but achieves significant H@100 improvement. In addition, for the inference speed under the PredCls task settings, our HMPNet can achieve 5.6 FPS on the A800 GPU device, which is 1.37× and 1.65× faster than the recently published methods HiKER (4.1 FPS) and SQUAT (3.4 FPS), respectively. In addition, the code base we just build has a lot of room for streamlining, and we will continue to purify it in the future to make the HMPNet model more efficient. Overall, HMPNet reaches the best tradeoff between accuracy and efficiency.

E. Ablation Studies

1) Effectiveness of SPO: As stated in Section III-C3, the SPO is introduced to align and compress the initial image and text subprototypes with heterogeneous information. We remove the SPO strategy (denoted by Ours + MHL) to verify its effectiveness and the results are reported in Table II. Its accuracy surpasses the baseline across all indicators. However, compared with the complete HMPNet, without the SPO strategy to constrain multimodal subprototypes, there is a 1.3% drop of mR@100 on the PredCls task because the intermodality gap of the same predicate and the intramodality semantic overlap of different predicates make the whole learning more difficult. Besides, the model without SPO reduces the H@100 value by 0.2% on the SGCls task due to the reduced resistance to existing noise in the predicted entity labels. Hence, the SPO strategy is necessary to help multimodal subprototypes enhance discriminability and robustness through more compact and consistent semantics.

2) Effectiveness of Multimodal Hypergraph Learning: The multimodal hypergraph is designed to mine the relevant contents among multiple relation predicates and capture complementary contextual information across modalities. To test the effectiveness of this module, we mask the multimodal hypergraph to observe the changes in model performance. As shown in Table II, after removing the MHL module, the performance of "Ours + SPO" is still stronger than the baseline model. But compared with the complete HMPNet, it decreases on all three subtasks (e.g., from 48.9% to 46.1% on mR@20 of PredCls, from 54.0% to 52.8% on mR@100 of SGCls, and from 17.7% to 16.9% on H@20 of SGDet) due to the suboptimal interaction of subprototypes. In addition, for the PredCls task, a decrease of 1.8% and 1.2% is observed on H@20 and H@100, respectively. This reveals that there is competition between heterogeneous modal information, and modality-independent prototypes will lead to confusion in the matching classification, thus reducing the accuracy of the overall relation predictions. Therefore, the multimodal hypergraph successfully solves the high-order correlation optimization problem and improves the reliability of the relation prediction by making the subprototypes of the two modalities more reasonable and precise.

3) Effect of the Number of Neighbors: The number of selected neighbors is a key hyperparameter in constructing the multimodal hypergraph's topological structure. It determines the degrees of hyperedges and affects high-order message passing. We set six different neighbors to explore the sensitivity to the number k of the model and the results of the PredCls, SGCls, and SGDet tasks are shown in Table III. When k is set to 1, each hyperedge contains one vertex, and the cross-modal hyperedge-level updating performs a cross-attention operation on all vertices (similar to a transformer block). This leads to a drop of 0.5% on H@100 in PredCls because the global subprototype vertices distract the attention. When k is set to 2, the performance decreases by 1.5% on mR@20 in PredCls because the hypergraph degenerates into a normal graph where position-free feature updates along edges are more likely to lead to smooth subprototype embeddings. After this, for the PredCls task, as k gradually increases from 3 to 6, the performance mR@100 of the model decreases from 76.1% to 74.9%. This demonstrates that three neighbors can well explore the relevant semantics among predicates, and more neighbors will lead to excessive transmission of messages and weaken the discriminative power of multimodal prototypes. For the SGCls and SGDet tasks, under different k value

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TABLE II

ABLATION STUDY ON EACH COMPONENT OF THE PROPOSED HMPNET. "SPO" DENOTES THE SUBPROTOTYPE OPTIMIZATION FOR TEXT AND IMAGE MODALITIES. "MHL" DENOTES THE MULTIMODAL HYPERGRAPH LEARNING MODULE

Component	PredCls			SGCls			SGDet		
	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100
Baseline (PE-Net [48])	61.8 / 87.6	45.6 / 71.4	52.5 / 78.7	52.5 / 71.8	28.7 / 48.4	37.1 / 57.8	27.8 / 38.2	11.9 / 19.9	16.7 / 26.2
Ours+MHL	62.6 / 88.5	47.5 / 74.8	54.0 / 81.1	53.8 / 73.9	31.0 / 53.7	39.3 / 62.2	27.8 / 38.3	12.7 / 21.4	17.4 / 27.5
Ours+SPO	62.6 / 88.4	46.1 / 74.3	53.1 / 80.7	53.1 / 73.3	30.8 / 52.8	39.0 / 61.4	27.8 / 38.2	12.1 / 21.2	16.9 / 27.3
Ours+SPO+MHL	**62.7 / 88.6**	**48.9 / 76.1**	**54.9 / 81.9**	**53.8 / 73.9**	**31.9 / 54.0**	**40.1 / 62.4**	**27.8 / 38.3**	**13.0 / 21.9**	**17.7 / 27.9**

TABLE III

EFFECT OF THE NUMBER OF THE SELECTED NEIGHBORS WHEN CONSTRUCTING THE MULTIMODAL HYPERGRAPH. SIX DIFFERENT k VALUES ARE SET TO EXPLORE THE OPTIMAL PARAMETER. THE RESULTS OF THE PREDCLS, SGCLS, AND SGDET TASKS ARE REPORTED IN DETAIL

Neighbors	PredCls			SGCls			SGDet		
k	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100
k=1	62.6 / 88.5	47.6 / 75.4	54.1 / 81.4	53.7 / 73.5	31.7 / 53.9	39.8 / 62.2	27.7 / 38.3	12.5 / 20.8	17.2 / 27.0
k=2	62.7 / 88.5	47.4 / 74.9	54.0 / 81.1	53.8 / 73.7	31.7 / 53.7	39.9 / 62.1	27.8 / 38.3	12.5 / 21.7	17.2 / 27.7
k=3	**62.7 / 88.6**	**48.9 / 76.1**	**54.9 / 81.9**	53.8 / 73.9	**31.9 / 54.0**	**40.1 / 62.4**	**27.8 / 38.3**	**13.0 / 21.9**	**17.7 / 27.9**
k=4	62.6 / 88.5	48.3 / 75.5	54.5 / 81.5	**54.0 / 74.1**	31.7 / 53.7	39.9 / 62.3	27.8 / 38.3	12.6 / 21.7	17.3 / 27.7
k=5	62.6 / 88.5	48.0 / 75.2	54.3 / 81.3	54.0 / 74.0	31.5 / 53.6	39.8 / 62.2	27.8 / 38.3	12.5 / 21.6	17.2 / 27.6
k=6	62.5 / 88.4	47.1 / 74.9	53.7 / 81.0	53.7 / 73.8	31.3 / 53.5	39.5 / 62.0	27.7 / 38.2	12.5 / 20.9	17.2 / 27.0

TABLE IV

EFFECT OF DIFFERENT POSITIONS OF THE SUBPROTOTYPE OPTIMIZATION STRATEGY. "PRE-SPO" DENOTES THAT SPO IS ADOPTED BEFORE THE HYPERGRAPH MODULE AND "POST-SPO" DENOTES THAT SPO IS PLACED AFTER THE HYPERGRAPH MODULE

Position	PredCls			SGCls			SGDet		
	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100
Post-SPO	62.6 / 88.5	48.2 / 75.4	54.5 / 81.4	53.7 / 73.9	31.4 / 53.6	39.6 / 62.1	27.8 / 38.3	12.3 / 21.5	17.1 / 27.5
Pre-SPO	**62.7 / 88.6**	**48.9 / 76.1**	**54.9 / 81.9**	**53.8 / 73.9**	**31.9 / 54.0**	**40.1 / 62.4**	**27.8 / 38.3**	**13.0 / 21.9**	**17.7 / 27.9**

settings, the overall change in model accuracy is consistent with the relevant analysis of the PredCls task above. As a pleasing explored result, when the k value is equal to 3, the model yields the best performance in most indicators across all tasks, so we set the number of selected neighbors to 3 in all experiments.

4) Effect of the Position of the SPO: The SPO is designed for the image and text subprototypes. The SPO is applicable in two places of our model, namely before and after the multimodal hypergraph module, which is denoted as Pre-SPO and Post-SPO, respectively. We train HMPNet using Pre-SPO as well as Post-SPO and the results are shown in Table IV. For the overall Recall, there is little difference (0.1% on R@100 in PredCls) between the two strategies. But for the mean Recall, the Pre-SPO brings improvements of 0.7% on mR@100 in PredCls, 0.4% on mR@100 in SGCls, and 0.7% on mR@20 in SGDet. This indicates that the image and text subprototypes should be optimized first and then fed into the hypergraph module, which results in better performance. Hence, the initial image and text subprototypes may contain some noise or ambiguous contexts and Pre-SPO can help make them more compact and precise. Then, the multimodal hypergraph module can achieve better reasoning results with the higher quality source vertices.

5) Different Weight Types of the Image Subprototype: The image subprototype is a key part of the proposed framework. Accordingly, we test various methods to construct image subprototype based on o_i (see Section III-C2), and the results of PredCls, SGCls, and SGDet tasks are summarized in Table V.

a) Extra embedding: In this case, the weights of image subprototypes are from predefined embeddings. After training o_i for some epochs, we initialize these embeddings using values of o_i and optimize them as image subprototypes. Intuitively, this approach makes the image subprototype more expressive using more parameters, but it is not perfect in terms of the mR@20/100 indicator because the early o_i may contain more noise and the supervision of ISP is relatively loose.

b) Copy weights: At each training iteration, we directly copy weights of o_i to produce the image subprototype. Thus, higher-quality o_i also promotes the optimization of other parts such as SPO, MHL, and matching classification. Compared with the above "extra embedding" case, it brings a rise of 0.5% on H@20 in PredCls but there is a drop of 0.1% on H@20 in SGDet. Better image subprototypes can be obtained and potential room for improvement exists in this case.

c) Share weights: To jointly optimize o_i and multimodal prototypes, we make image subprototypes share weights with o_i. That is to say, not only the weight of o_i is inherited, but also the gradient of the multimodal hypergraph is passed to o_i during training. Table V shows that this case can lead to the best comprehensive performance on H@20/100, i.e., 54.9%/81.9% in PredCls, 40.1%/62.4% in SGCls, and

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TABLE V
EFFECT OF DIFFERENT METHODS FOR INITIALIZING AND OPTIMIZING IMAGE SUBPROTOTYPE

Type	PredCls			SGCls			SGDet		
	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100
Extra Embedding	62.6 / 88.5	47.5 / 75.5	54.0 / 81.5	53.8 / 73.8	31.8 / 53.9	40.0 / 62.3	27.8 / 38.3	12.8 / 21.6	17.5 / 27.6
Copy Weights	62.6 / 88.5	48.2 / 75.7	54.5 / 81.6	53.8 / 73.7	31.6 / 53.8	39.8 / 62.2	27.7 / 38.2	12.7 / 21.4	17.4 / 27.4
Share Weights	**62.7 / 88.6**	**48.9 / 76.1**	**54.9 / 81.9**	**53.8 / 73.9**	**31.9 / 54.0**	**40.1 / 62.4**	**27.8 / 38.3**	**13.0 / 21.9**	**17.7 / 27.9**

TABLE VI
EXPLORATION FOR THE MULTIMODAL CALIBRATION STRATEGY. THE EFFECTS OF SEVERAL STRATEGIES ARE REPORTED WHEN CALCULATING RELATION SCORES VIA REPRESENTATION-PROTOTYPE MATCHING LOGITS DURING THE INFERENCE STAGE

Strategy	PredCls			SGCls			SGDet		
Detail	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100	R@20/100	mR@20/100	H@20/100
TSP	62.7 / 88.5	46.0 / 74.8	53.1 / 81.1	53.8 / 73.8	30.4 / 52.8	38.8 / 61.6	27.7 / 38.3	12.4 / 21.5	17.1 / 27.5
TSP+ISP	62.7 / 88.6	47.7 / 75.5	54.2 / 81.5	53.8 / 73.7	31.6 / 53.9	39.8 / 62.3	27.6 / 38.2	12.5 / 21.8	17.2 / 27.8
func(TSP, ISP)	62.7 / 88.5	47.7 / 75.3	54.2 / 81.4	51.8 / 70.7	28.4 / 48.6	36.7 / 57.6	25.8 / 33.6	12.1 / 17.5	16.5 / 23.0
Pre-ISP	62.3 / 88.3	44.6 / 72.1	52.0 / 79.4	53.5 / 73.7	30.6 / 51.4	38.9 / 60.6	27.7 / 38.3	12.5 / 21.3	17.2 / 27.4
CW(TSP, ISP)	62.7 / 88.6	46.0 / 75.0	53.1 / 81.2	53.8 / 73.8	31.5 / 53.7	39.7 / 62.2	27.7 / 38.3	12.3 / 20.7	17.0 / 26.9
ISP	62.7 / 88.6	48.8 / 75.9	54.9 / 81.8	53.8 / 73.8	31.8 / 53.9	40.0 / 62.3	27.8 / 38.3	12.9 / 21.9	17.6 / 27.9
RC(TSP, ISP)	**62.7 / 88.6**	**48.9 / 76.1**	**54.9 / 81.9**	**53.8 / 73.9**	**31.9 / 54.0**	**40.1 / 62.4**	**27.8 / 38.3**	**13.0 / 21.9**	**17.7 / 27.9**

17.7%/27.9% in SGDet. As a result, we adopt this setting to produce image subprototype in all experiments.

6) Exploration for Multimodal Calibration Strategy: The multimodal calibration strategy is devised to boost the accuracy by achieving modality-balance representation-prototype matching. During training, two modalities are used to calibrate the distribution of logits in the matching process. During inference, based on the ISP-based and TSP-based matching logits, the subsequent calculating strategy of the predicted relation score is crucial to the final performance. We investigate multiple strategies and the results are shown in Table VI.

a) TSP: As a straight attempt, we use the TSP-based matching logits as the predicted score of each relation sample and obtain the relation category by the following formula:

$$c_{\text{res}} = \arg\max_k \delta\left(\left\{q_k \middle| q_k = \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^t} \right\rangle\right\}_{k=1}^{C_r}\right) \tag{45}$$

where c_{res} is the class index of the final result. r_{ij} denotes a relation representation of the testing sample to be predicted and $\widetilde{p}_k^t$ is the updated text subprototype. On the PredCls task, this strategy yields the R@100 of 88.5% but the mR@100 of 74.8% is not satisfactory due to the upper limit of supervision.

b) TSP + ISP: We add the TSP-based and ISP-based matching logits together and perform the softmax δ to get the final output scores, which can be expressed as follows:

$$c_{\text{res}} = \arg\max_k \delta\left(\left\{q_k \middle| q_k = \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^t} \right\rangle + \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^v} \right\rangle\right\}_{k=1}^{C_r}\right) \tag{46}$$

where $\widetilde{p}_k^v$ is the updated image subprototype and C_r is the total number of relation categories. With this strategy, the model produces the decent mR@20 of 47.7% on the PredCls task and mR@20 of 31.6% on the SGCls task. This indicates that the image subprototypes and text subprototypes are aligned to some extent and can work together to lift the accuracy ceiling of the text modality.

c) func(TSP, ISP): In addition to the simple method above, we consider a learnable weighted combination approach to find a tradeoff between TSP-based and ISP-based logits. The specific formulas are as follows:

$$s_k^t = \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^t} \right\rangle, \quad s_k^v = \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^v} \right\rangle \tag{47}$$

$$c_{\text{res}} = \arg\max_k \delta\left(\left\{q_k \middle| q_k = \alpha \cdot s_k^t + \beta \cdot s_k^v\right\}_{k=1}^{C_r}\right) \tag{48}$$

where α and β are the trainable factors to control the proportion of the two logit parts. It can be seen that this strategy performs well on the PredCls task. However, it does not meet the expectations on the complex SGCls and SGDet tasks and yields the lowest H@100 values of 57.6% and 23.0%. This indicates that it is more difficult to train this weighted strategy when the given entity labels and bounding boxes are noisy under the settings of SGCls and SGDet tasks.

d) Pre-ISP: We can also employ the initial visual relation classifier [i.e., embedding o_i in (10)] to perform the final relation recognition, which can be formulated as follows:

$$c_{\text{res}} = \arg\max_k \delta\left(\left\{q_k \middle| q_k = \left\langle \overline{r_{ij}}, \overline{o_k} \right\rangle\right\}_{k=1}^{C_r}\right). \tag{49}$$

In this case, the model yields a poor 44.6%/72.1% of mR@20/100 on the PredCls task. This is reasonable because the weights of the naive visual classifier do not incorporate textual information and produce suboptimal results.

e) CW (TSP, ISP): In this case, we try to train a more fine-grained coefficient for each relation category to fuse TSP and ISP. We employ an extra MLP layer W_{cw} to generate a class-specific weight vector $\mathbf{a}_{\text{cw}}$

$$\mathbf{a}_{\text{cw}} = \sigma\left(W_{\text{cw}}\left[S^t; S^v\right]\right) \tag{50}$$

$$c_{\text{res}} = \arg\max \delta\left(\mathbf{a}_{\text{cw}} S^t + (1 - \mathbf{a}_{\text{cw}}) S^v\right) \tag{51}$$

where σ is the sigmoid function and S^t/S^v are the TSP-based and ISP-based matching logits. δ is the softmax function. On the three SGG tasks, this strategy yields decent R@20/100,

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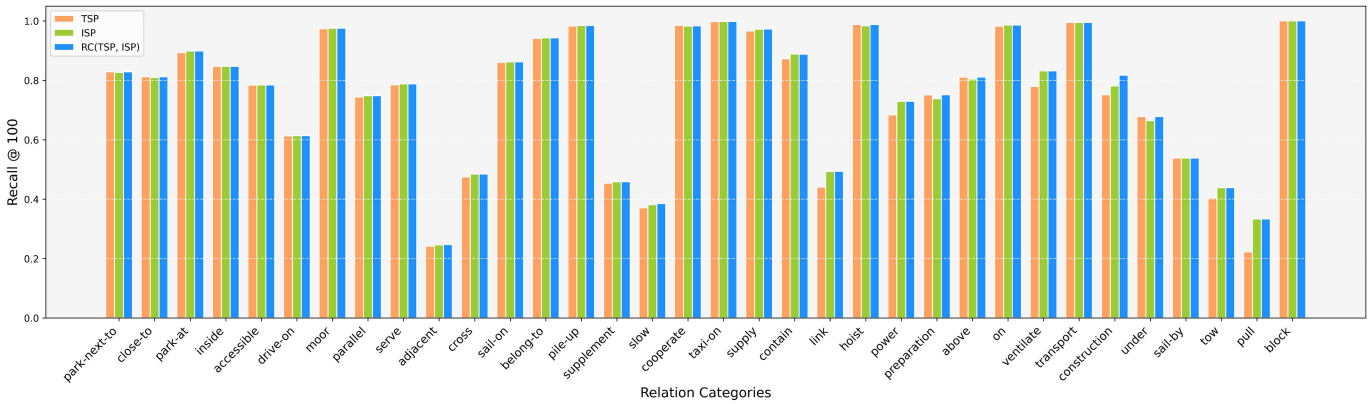


Fig. 10. Comparison of different multimodal calibration strategies on the PredCls task. After modality-balance calibration during training, the ISP and ISP strategies show similar testing distributions. Moreover, the RC(TSP, ISP) strategy can organically combine the advantages of both to some extent.

but performs poorly on mR@20/100. This indicates that the class-specific weight vector may cause error accumulation in some hard predicate categories when combining both matching logits.

f) ISP: For this point, we adopt the ISP-based matching logits as the predicted scores to output the relation category

$$c_{\text{res}} = \arg\max_{k} \delta\left(\left\{q_k \middle| q_k = \left\langle \overline{r_{ij}}, \overline{\widetilde{p}_k^v} \right\rangle\right\}_{k=1}^{C_r}\right). \tag{52}$$

As shown in Table VI, suppose we ignore the refined RC(TSP, ISP), and this ISP strategy without any bells and whistles reaches the best H@20/100 of 54.9%/81.8%, 40.0%/62.3%, and 17.6%/27.9% on all three challenging tasks. This demonstrates that through SPO and hypergraph reasoning, image subprototypes are purified after obtaining complementary semantics from the textual modality. Moreover, with sufficient supervision of the initial relation visual classifier and the multimodal calibration strategy, text-guided image subprototypes enjoy accurate referential semantics and strong robustness.

g) RC(TSP, ISP): As shown in (43), we propose the adaptive confidence to combine both matching logits. In Table VI, the refinement calibration strategy works effectively during inference and yields the best performance on the PredCls, SGCls, and SGDet tasks. In addition, Fig. 10 shows more clearly the effect of the RC(TSP, ISP) strategy on each predicate category. By comprehensively evaluating the maximum matching score and information entropy, we can select a more reliable ISP or TSP as the prediction result for each relationship sample. The performance improvements of some predicates meet our expectations, e.g., "construction" (abbreviation for "support the construction of"), "under" and "pull". We believe this strategy will yield improved performance as the quality of multimodal prototypes is enhanced, and we hope it will inspire further research.

Hence, we equip HMPNet with the most effective RC(TSP, ISP) strategy, which is the same as (43).

The above results show that the relation prediction ceiling can be lifted by combining the rich details of the image modality and the high-level semantics of the text modality during the training stage. This is an important idea for promoting remote sensing scene understanding tasks. The proposed hypergraph-guided multimodal prototypes and the multimodal calibration strategy are successful and convincing practices.

V. Conclusion

In this article, we propose a novel HMPNet to predict the relations between target pairs, promoting remote sensing scene understanding by capturing spatial and semantic associations. We devise the multimodal calibration strategy to enhance the discriminability and make the prototype-representation matching process more balanced. Moreover, the multimodal hypergraph is devised to efficiently learn the relevant hybrid semantic contents across the multimodal subprototypes. As a result, the proposed HMPNet can showcase competitive performance on the remote sensing SGG task. We encourage more researchers to pay more attention to the remote sensing scene understanding beyond the entity-level perception.

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